Expectation Propagation Notes

Joaquín Rapela

February 1, 2026

These notes were mostly extracted from Bishop (2016)

1 Fundamental result

Consider the problem of minimizing $\mathrm{KL}(p||q)$ with respect to $q(\mathbf{z})$, when $p(\mathbf{z})$ is a fixed distribution, and $q(\mathbf{z})$ is a member of the exponential family:

$$q(\mathbf{z}) = h(\mathbf{z})g(\eta)\exp(\eta^\intercal \mathbf{u}(\mathbf{z}))$$

The minimizing distribution $q(\mathbf{z})$ satisfies the moment matching criterion:

$$\mathbb{E}_{q(\mathbf{z})}\left[\mathbf{u}(\mathbf{z})\right] = \mathbb{E}_{p(\mathbf{z})}\left[\mathbf{u}(\mathbf{z})\right]$$

2 Summary of expectation propagation

We are given the joint probability density function between the data $\mathcal{D}$ and hidden variables and parameters θ:

$$p(\mathcal{D}, \theta) = \prod_i f_i(\theta)$$

For example, if we are given N independent and identically distributed data points $\{\mathbf{x}_1, \ldots, \mathbf{x}_N\}$, then $f_i(\theta) = p(\mathbf{x}_i|\theta)$, for $i \in \{1, \ldots, N\}$, and $f_0(\theta)$ is the prior on the hidden variables and parameters, $f_0(\theta) = p(\theta)$.

We want to estimate

$$p(\theta|\mathcal{D}) = \frac{1}{p(\mathcal{D})}\prod_i f_i(\theta) \quad \text{with} \quad p(\mathcal{D}) = \int \prod_i f_i(\theta) d\theta \tag{1}$$

We assume that the marginalization in Eq. 1 is intractable, and we will approximate this true posterior with

$$q(\theta) = \frac{1}{Z}\prod_i \tilde{f}_i(\theta)$$

where $\tilde{f}_i(\theta)$ are members of the exponential family.

We could find the factors $\tilde{f}_i(\theta)$ by minimizing

$$\mathrm{KL}\left(\frac{1}{p(\mathcal{D})}\prod_i f_i(\theta)\middle\|\frac{1}{Z}\prod_i \tilde{f}_i(\theta)\right)$$

However, this minimization is intractable, as we need to simultaneously fit all terms of the complex distribution $\frac{1}{p(\mathcal{D})}\prod_i f_i(\theta)$.

An alternative approach would fit separately the factors $f_i(\theta)$ and $\tilde{f}_i(\theta)$. But this approach could overfit some factors, by not taking into consideration the other factors.

Expectation propagation fits the term $\tilde{f}_i(\theta)$ to the term $f_i(\theta)$ in the context of previously fitted terms. It starts by initializing all factors $\tilde{f}_i(\theta)$ and then cycles each factor optimizing them one at a time in the context of other factors. Suppose it wants to optimize the factor $\tilde{f}_j(\theta)$. It starts by removing this factor from the product, giving the unnormalized distribution $q^{\setminus j}(\theta) = \prod_{i\neq j}\tilde{f}_i(\theta)$, and then estimates

$$\tilde{f}_j^{\mathrm{new}}(\theta) = \underset{\tilde{f}_j(\theta)}{\arg\min}\left\{\mathrm{KL}\left(\frac{1}{Z_j}f_j(\theta)\prod_{i\neq j}\tilde{f}_i(\theta)\middle\|\frac{1}{Z}\tilde{f}_j(\theta)\prod_{i\neq j}\tilde{f}_i(\theta)\right)\right\}$$

It does this estimation in two steps. First, it estimates $q^{\mathrm{new}}(\theta)$ as

$$q^{\mathrm{new}}(\theta) = \underset{q(\theta)}{\arg\min}\left\{\mathrm{KL}\left(\frac{1}{Z_j}f_j(\theta)\prod_{i\neq j}\tilde{f}_i(\theta)\middle\|q(\theta)\right)\right\}$$

and second it estimates $\tilde{f}_j^{\mathrm{new}}(\theta) = K\frac{q^{\mathrm{new}}(\theta)}{q^{\setminus j}(\theta)}$. Because $q(\theta)$ is from the exponential family, from Section 1, the mean parameters μ of $q^{\mathrm{new}}(\theta)$ are $\mu = E_{q(\theta)}[u(\theta)] = E_{\frac{1}{Z_j}f_j(\theta)q^{\setminus j}(\theta)}[u(\theta)]$. This is called moment matching. Thus, $q^{\mathrm{new}}(\theta)$ is obtained by taking $E_{\frac{1}{Z_j}f_j(\theta)q^{\setminus j}(\theta)}[u(\theta)]$ as mean parameters. Expectation propagation assumes that the last expectation is tractable. For example, if $q(\theta)$ is a Normal distribution, then its mean and covariance are the mean and covariance of $\frac{1}{Z_j}f_j(\theta)q^{\setminus j}(\theta)$.

From the expression of $\tilde{f}_j^{\mathrm{new}}(\theta)$ in the proceeding paragraph we get $K = \int \tilde{f}_j^{\mathrm{new}}(\theta)q^{\setminus j}(\theta)d\theta$. Thus, K can be approximated by zeroth-order moment matching; i.e., $K = \int \tilde{f}_j^{\mathrm{new}}(\theta)q^{\setminus j}(\theta)d\theta \simeq \int f_j(\theta)q^{\setminus j}(\theta)d\theta$.

3 Pseudo code and Python implementation

Pseudo code for the expectation propagation algorithm is provided in Algorithm 1, and Python code implementing this algorithm can be found here.

References

Bishop, C. M. (2016). *Pattern recognition and machine learning.* Springer-Verlag New York.

Algorithm 1 Expectation propagation

We are given a joint distribution over observed data $\mathcal{D}$ and latent variables θ in the form of a product of factors

$$p(\mathcal{D}, \theta) = \prod_i f_i(\theta)$$

and we wish to approximate the posterior distribution $p(\theta|\mathcal{D})$ by a distribution $q(\theta)$ in the exponential family. We also wish to approximate the model evidence $p(\mathcal{D})$.

1. Initialize all of the approximating factors $\tilde{f}_i(\theta)$.
2. Initialize the posterior approximation $q(\theta) \propto \prod_i \tilde{f}_i(\theta)$.
3. Repeat until convergence:

   (a) Choose a factor $f_i(\theta)$ to be refined.

   (b) Remove the current approximating factor $\tilde{f}_i(\theta)$ from the posterior by division:

$$q_{\setminus i}(\theta) = \frac{q(\theta)}{\tilde{f}_i(\theta)}$$

   (c) Evaluate the new posterior $q^{\text{new}}(\theta)$ by matching the moments of

$$\frac{1}{Z_i} f_i(\theta) q_{\setminus i}(\theta)$$

   including evaluation of the normalization constant

$$Z_i = \int f_i(\theta) q_{\setminus i}(\theta) d\theta$$

   (d) Compute the revised factor

$$\tilde{f}_i(\theta) = Z_i \frac{q^{\text{new}}(\theta)}{q_{\setminus i}(\theta)}$$

4. The resulting approximation to the evidence is given by

$$p(\mathcal{D}) \simeq \int \prod_i \tilde{f}_i(\theta) d\theta$$